Karolinska Institutet

Cutting Through Resistance: Lessons from the EVA Trial on Episiotomy in Vacuum Birth

From skepticism to completion: navigating resistance and sustaining a seven-year randomized trial (and getting funding again)

Who am I?

Sophia Brismar Wendel
MD, PhD
Ob Gyn

2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026

The EVA trial

thebmj Research Education News & Views Campaigns Jobs

Research

Lateral episiotomy or no episiotomy in vacuum assisted delivery in nulliparous women (EVA): multicentre, open label, randomised controlled trial

BMJ 2024 ; 385 doi: https://doi.org/10.1136/bmj-2023-079014 (Published 17 June 2024)

PICO

Population: Nulliparous, vacuum birth, ≥34+0 weeks, singleton, live

Intervention: Lateral episiotomy

Comparison: No episiotomy

Outcome: Obstetric anal sphincter injury (OASI)

WHY DID WE DO THIS TRIAL?

BACKGROUND

AFTONBLADET

NYHETER SPORT NÖJE KULTUR LEDARE DEBATT TV PLUS SENAST

Aftonbladet granskar förlossningsskador

Det senaste om **Aftonbladet granskar förlossningsskador**. Läs nyheter, artiklar och se tv-klipp om **Aftonbladet granskar förlossningsskador** på Aftonbladet.se.

13 april AFTONBLADET TV

Barnmorskan förklarar: Så funkar det

– Vi fick allvarliga förlossningsskador

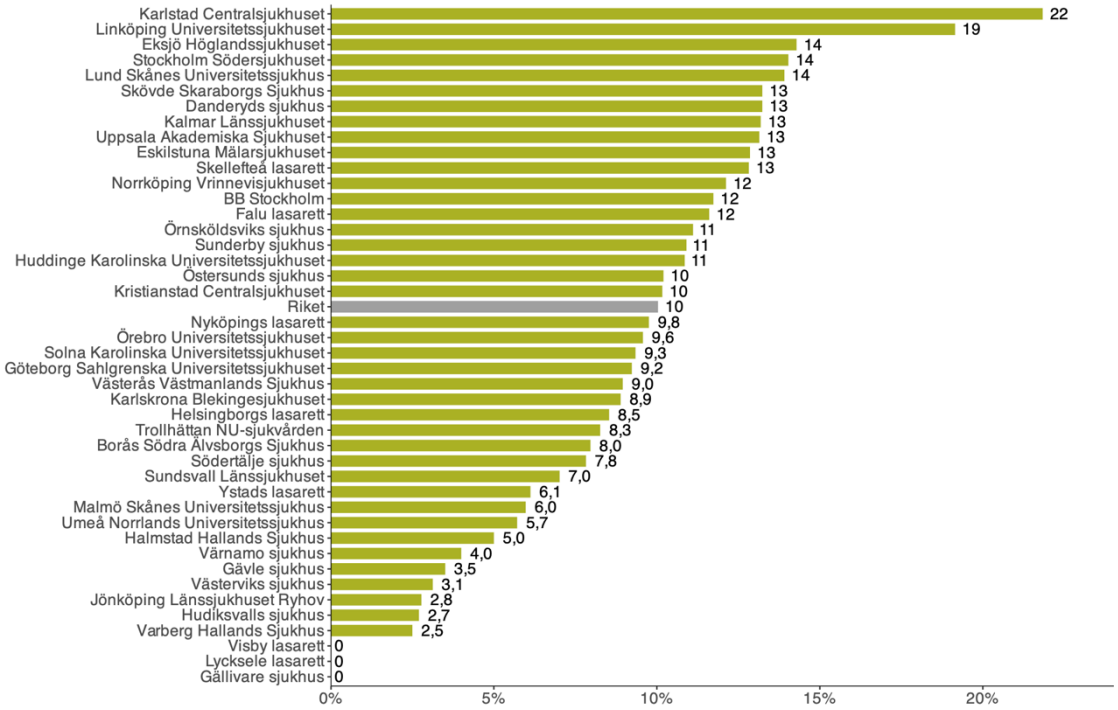
Fyra kvinnors egna berättelser ✔ Avslitna muskler ✔ Sfinkterruptur ✔ Opererats 22 gånge

Sugklocka ökar risken för bristningar

Men det kan rädda livet på barnet

Vacuum birth with episiotomy

OASI in vacuum birth – wide variation

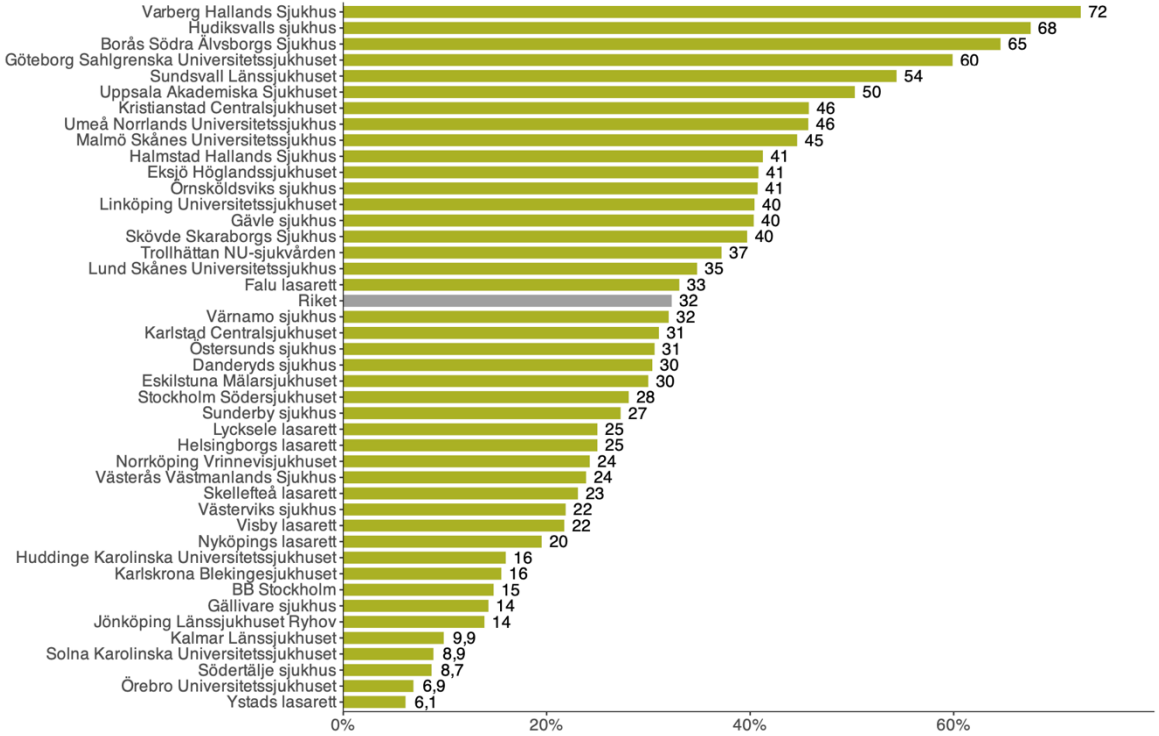


Max 22 %

Mean 10 %

Min 0 %

Episiotomy in vacuum birth – wide variation



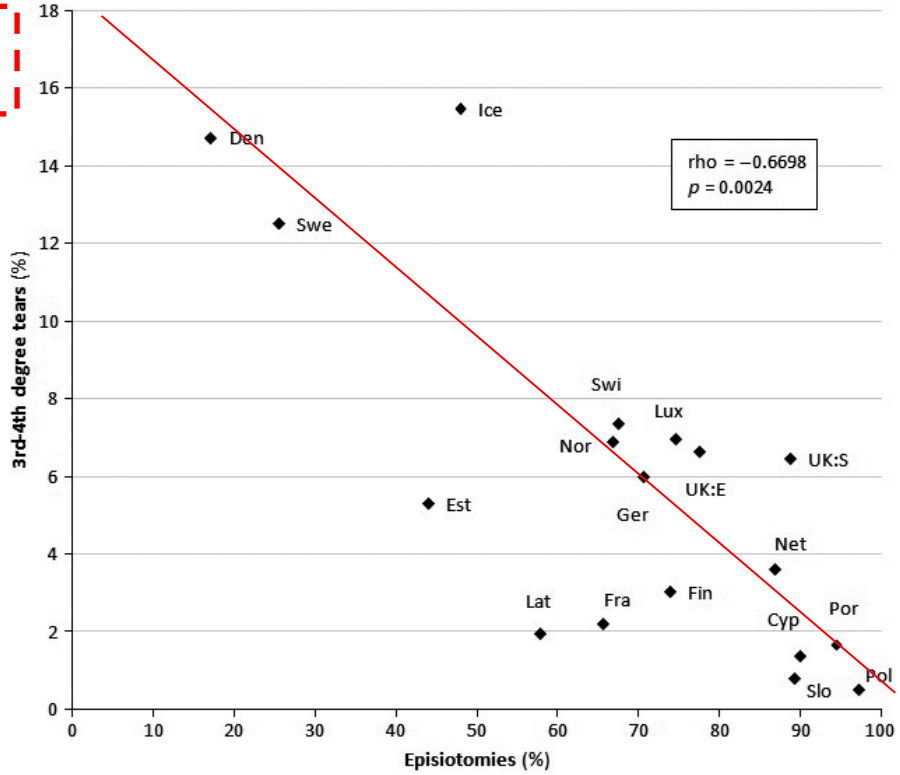
Max 72 %

Mean 32 %

Min 6 %

OASI and episiotomy in Europe 2010

Prevention problem?



Diagnostic problem?

Blondel et al, Acta Obstetricia et Gynecologica Scandinavica 2016

Metanalysis 2016 observational studies – episiotomy reduced OASI

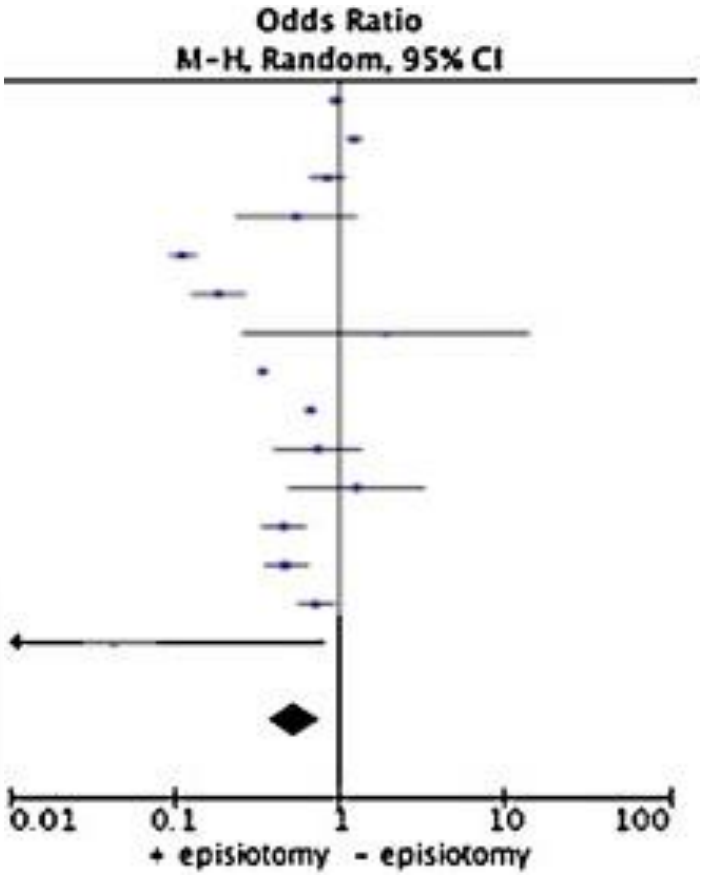
OR 0.53 (95% CI 0.37-0.77

NNT 18.3 (95% CI 17.7-18.9)

Review

Episiotomy in vacuum-assisted delivery affects the risk of obstetric anal sphincter injury: a systematic review and meta-analysis

Ninna S. Lund[a,*], Lisa K.G. Persson[b], Hanna Jangö[c], Ditte Gommesen[d], Hanne B. Westergaard[e]



HOW DID WE DO THIS TRIAL?

METHODS

2015

Ethical approval (2015/1238-31-2) with renewal for each site

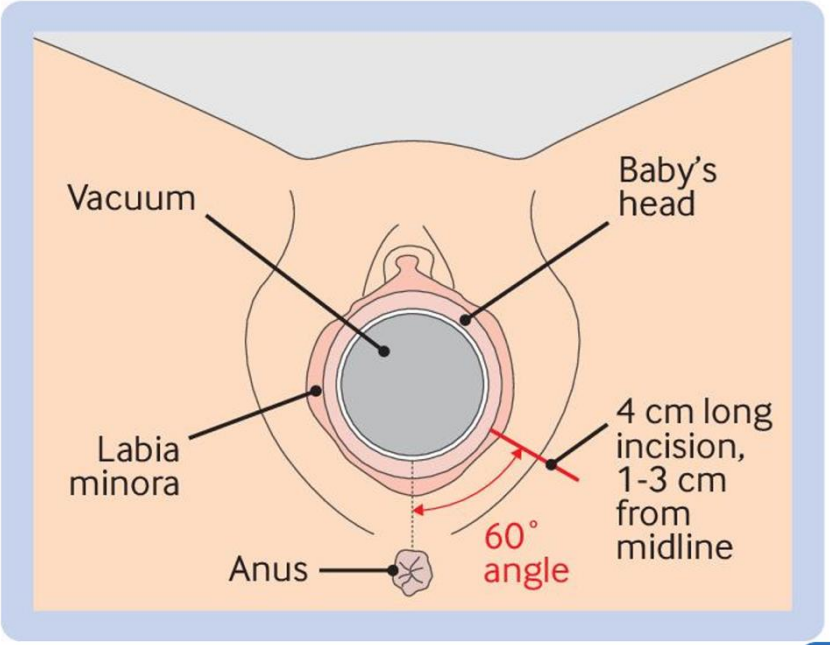
Registered in clinicaltrials.gov NCT02643108

2016

Swedish Research Council grant
12.9 MSEK 2017-2020

Supported by the Swedish network of clinical trials in obstetrics and gynecology

Intervention – lateral episiotomy



Sandra Bergendahl et al. BMJ 2024;385:bmj-2023-079014

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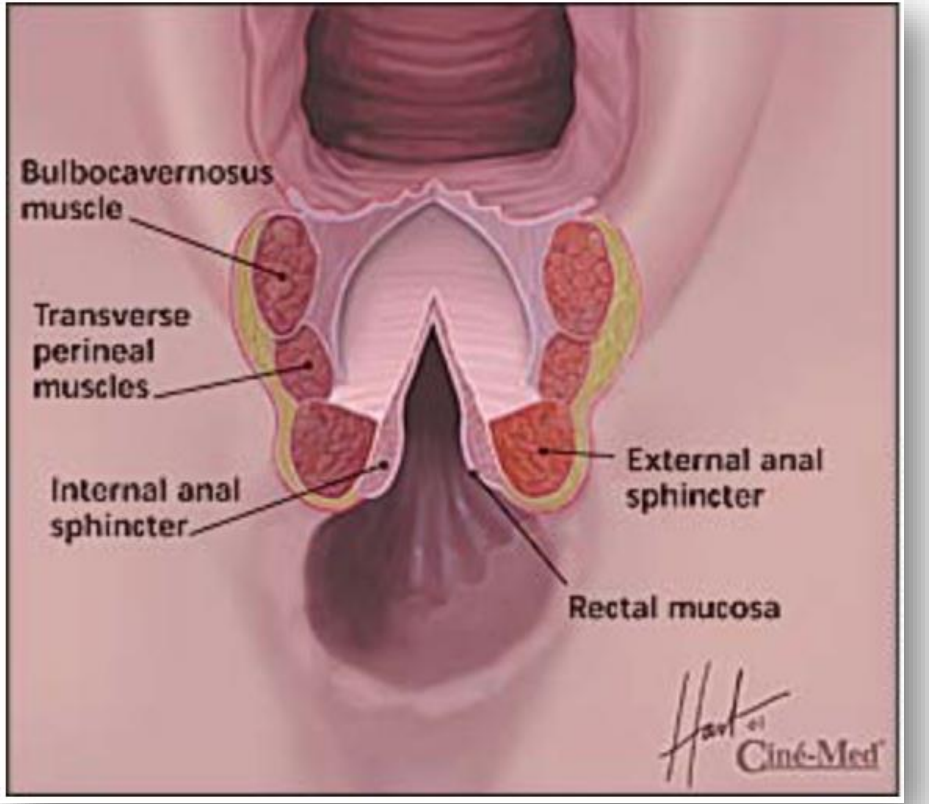
Outcome – obstetric anal sphincter injury



Study design

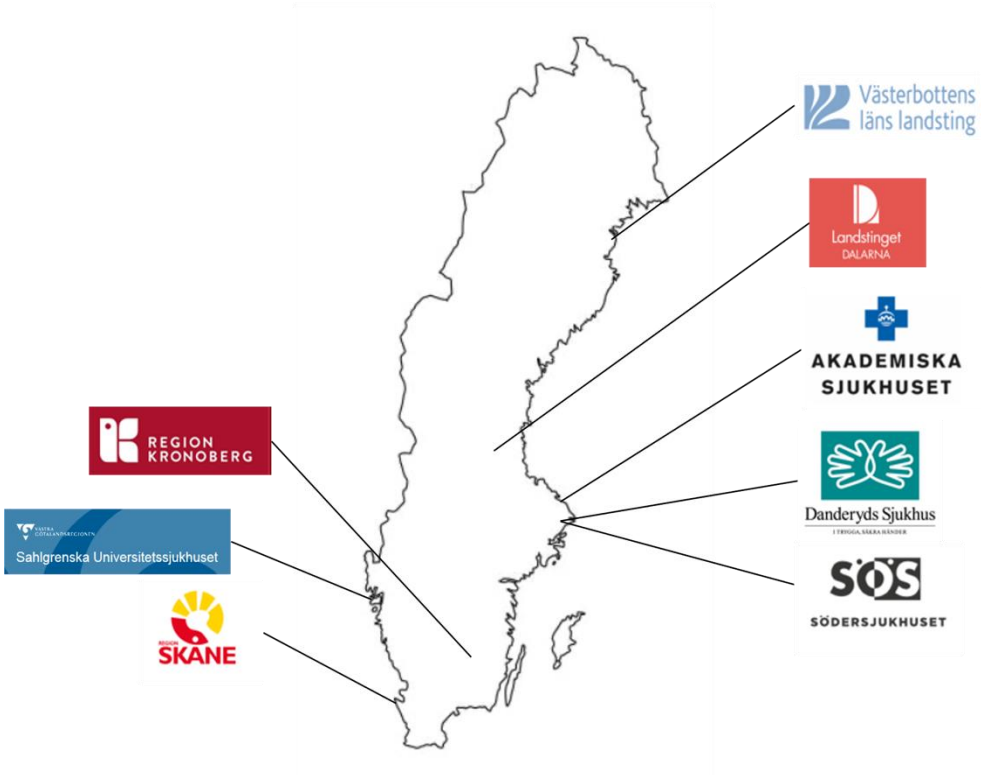
Multicenter RCT
8 hospitals
1 July 2017 to 15 February 2023

Inclusion criteria

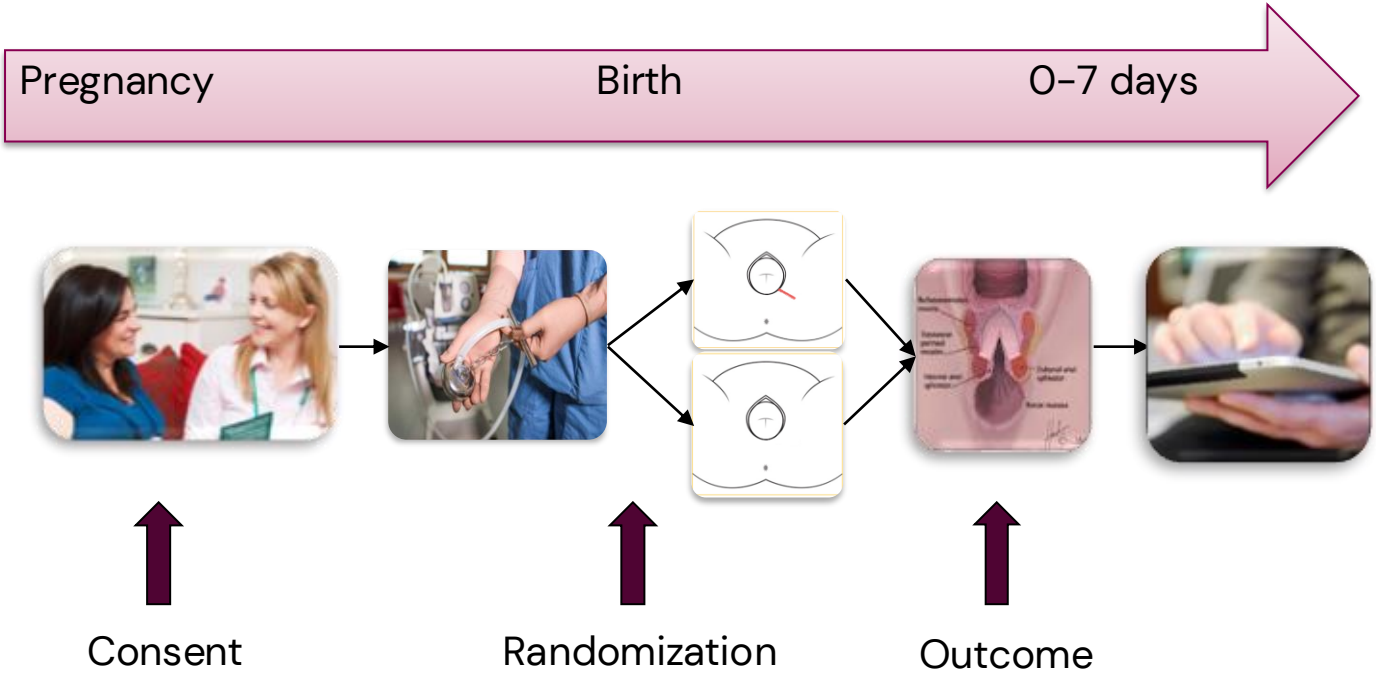
- Nulliparous, singleton, live fetus
- Week 34+0 or more
- Requiring vacuum

Exclusion criteria

- Previous surgery for incontinence or genital prolapse



EVA trial outline



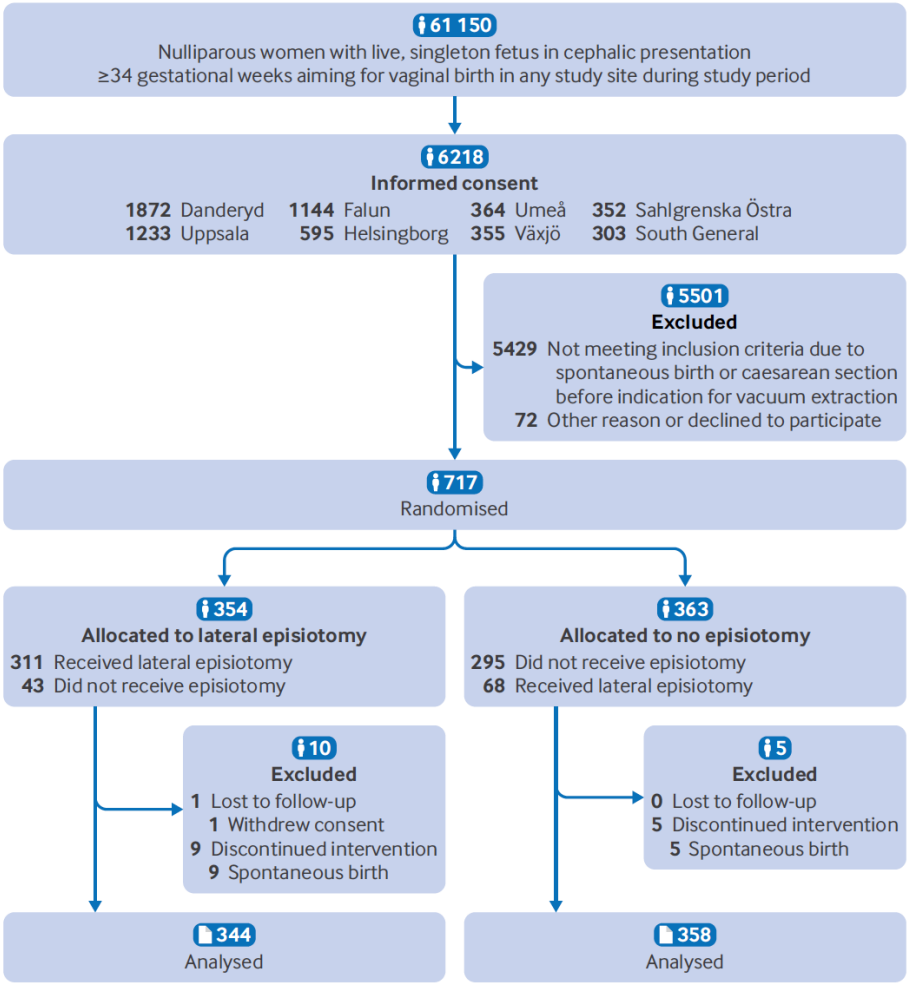
EVA trial follow-up postpartum

WHAT DID WE FIND IN THIS TRIAL?

RESULTS

Flowchart

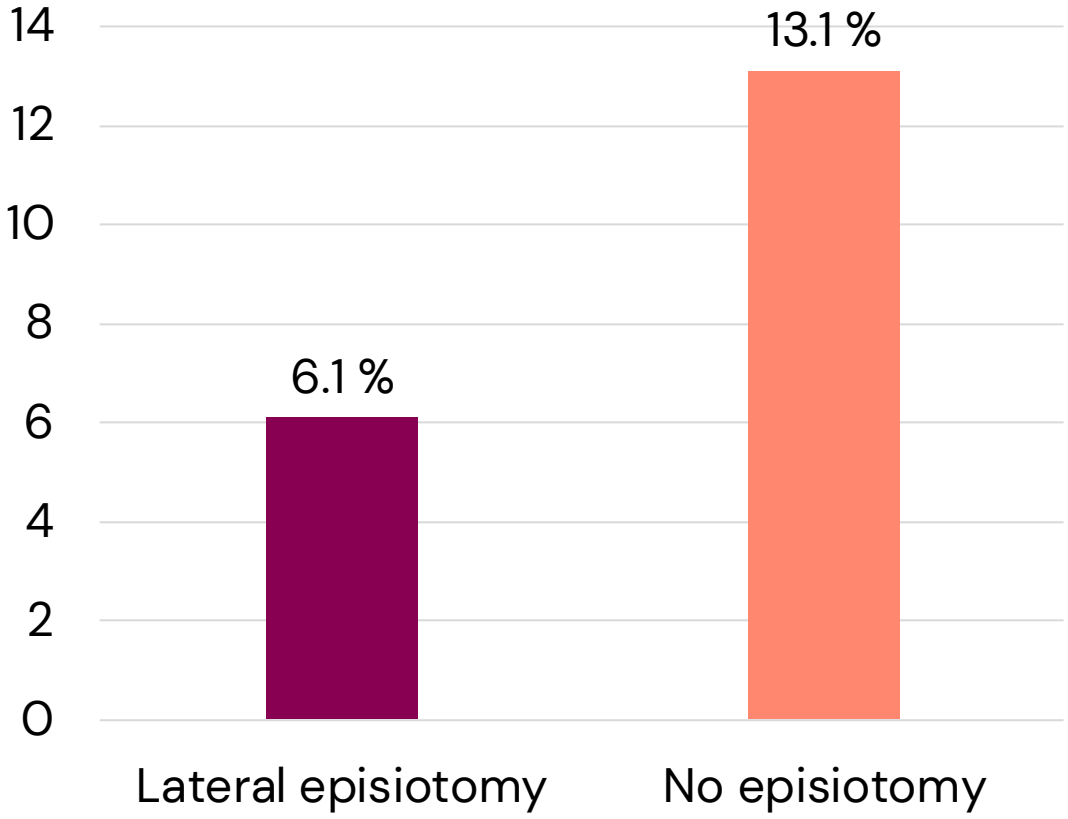
Final mITT population n=702



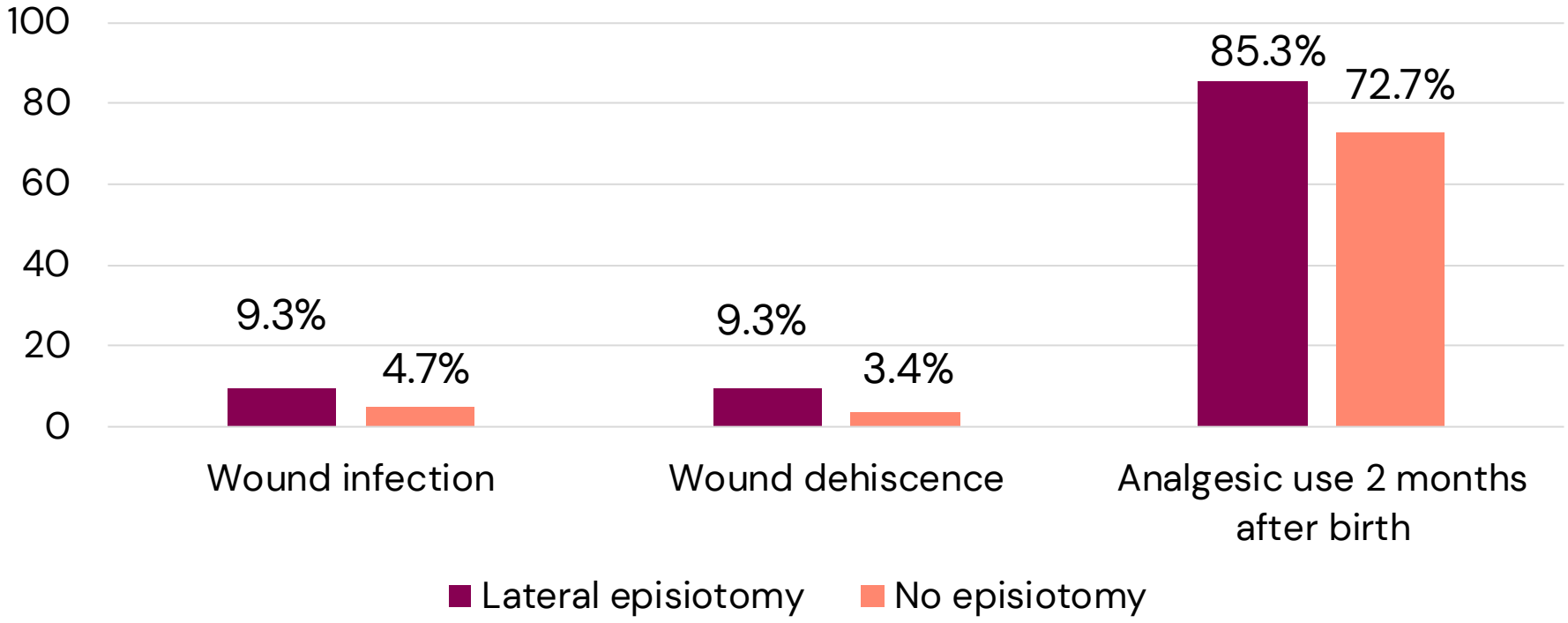
OASI decreased

RR 0.46 (0.28–0.78)

aRR 0.47 (0.23–0.97)



Some adverse events increased



WHAT DOES IT MEAN?

DISCUSSION

Strengths of this trial

- First adequately sized RCT for this research question
- Confirmed observational findings
- Excellent adherence to protocol
- Patient-reported and staff-reported outcomes

Limitations of this trial

- No masking – detection bias? Two examiners diagnosed tear
- Mostly Swedish women – selection bias?
- Only short-term follow-up (for now)

Conclusion

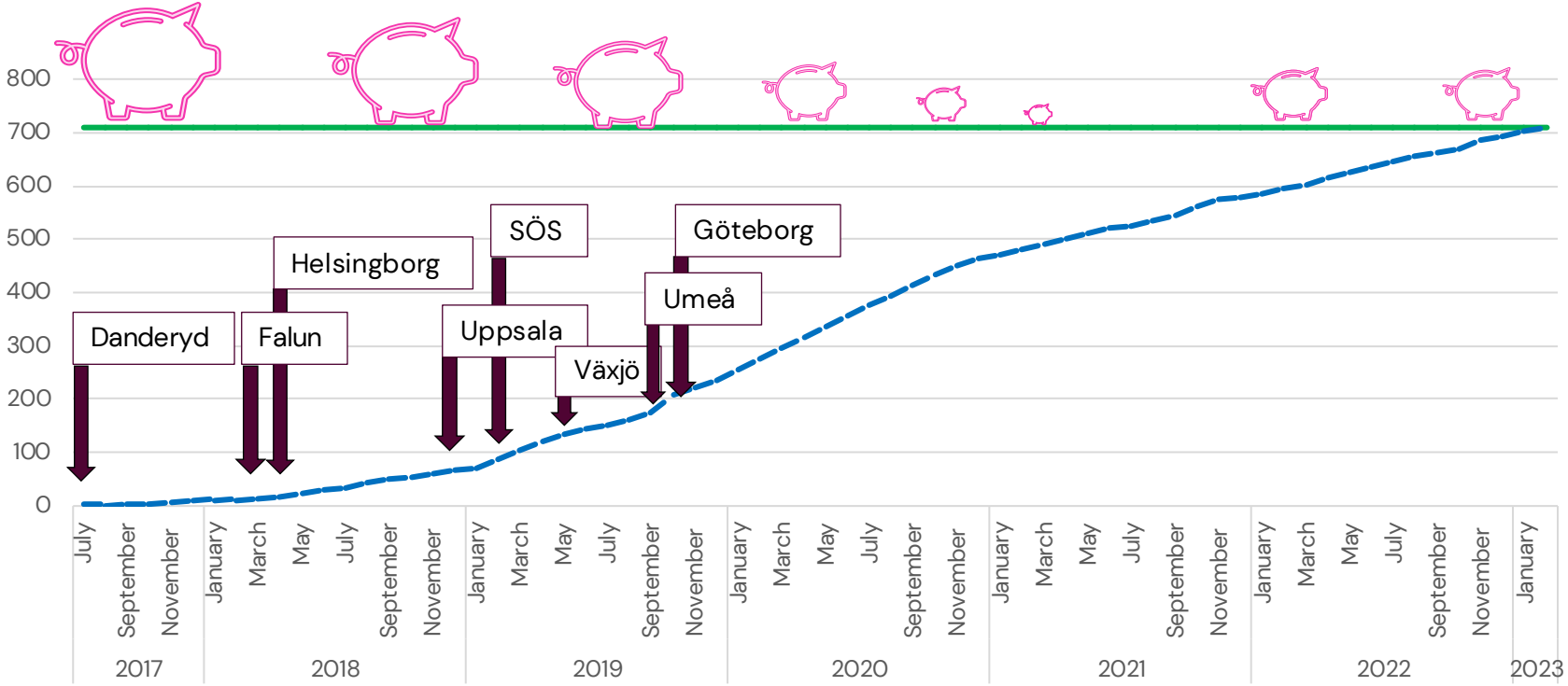
Our results confirm what has been reported in meta-analyses of observational studies.

Routine lateral episiotomy can be recommended for nulliparous women who require vacuum extraction to reduce OASI.

HOW DID WE OVERCOME THE RESISTANCE?

CHALLENGES

Recruitment



Informed consent
= a great challenge

Leaflets, home pages

Youtube film

Parental education

Personal information

– How do you do it?

Women's experiences of the consent process

Interviews with 23 women (mixed yes/no consent)

→ Most wanted information early

→ Motives were a good cause and better care

→ Typical fear: not getting an episiotomy if necessary

Ericson *et al. Trials* (2021) 22:658
https://doi.org/10.1186/s13063-021-05624-8

Trials

RESEARCH Open Access

Women's experiences of receiving information about and consenting or declining to participate in a randomized controlled trial involving episiotomy in vacuum-assisted delivery: a qualitative study

Jenny Ericson[1,2,3*], Cecilia Anagrius[4], Agnes Rygaard[4], Lisa Guntram[5], Sophia Brismar Wendel[6] and Susanne Hesselman[2,4,7]

Midwives' attitudes?

Midwives' experiences about the consent process

Interviews with 19 midwives

→ Understanding varied widely

→ Became more confident with time

Variation among staff important to consider and manage when relying on staff to recruit to an RCT

Sexual & Reproductive Healthcare 30 (2021) 100674

Contents lists available at ScienceDirect

Sexual & Reproductive Healthcare

journal homepage: www.elsevier.com/locate/srhc

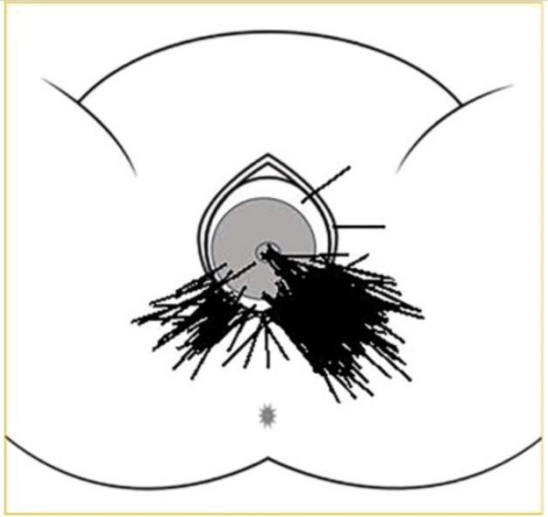
Episiotomy or not? Midwives' experiences of recruiting to a randomized controlled trial during pregnancy and labor – A qualitative study

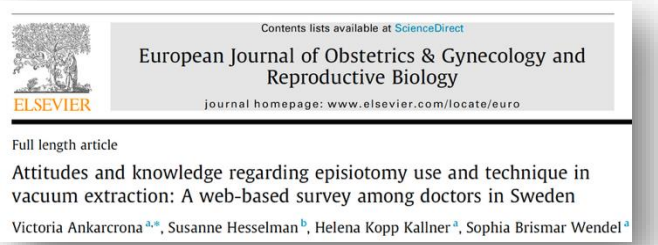
Mari-Cristin Malm [a,*], Susanne Hesselman [b,c], Sophia Brismar Wendel [d], Jenny Ericson [c,e,f]

Doctors' attitudes and knowledge

384 participants in a web questionnaire

- → 58% did episiotomy at <50% of vacuums
- → <50% could draw a correct episiotomy
- → Episiotomy was ranked "least important"
- → NNT ≤14 was required to do a routine episiotomy





Contents lists available at ScienceDirect

European Journal of Obstetrics & Gynecology and Reproductive Biology

journal homepage: www.elsevier.com/locate/euro

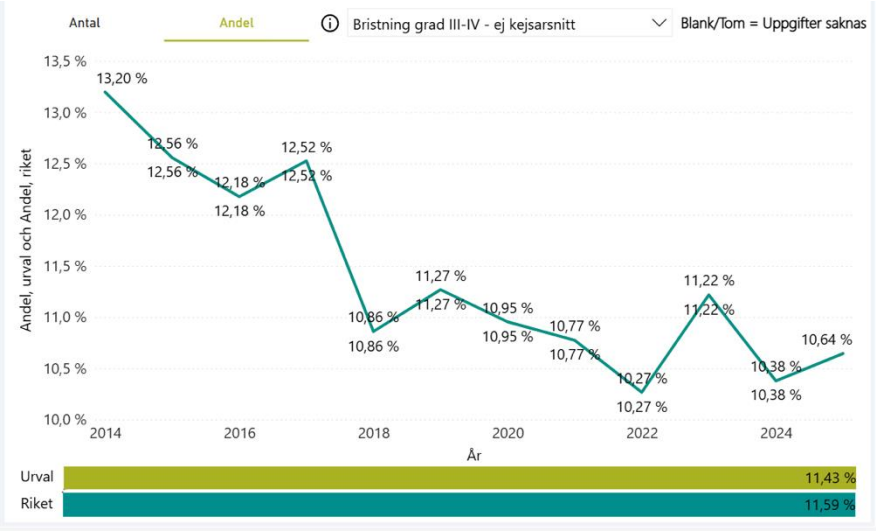
Full length article

Attitudes and knowledge regarding episiotomy use and technique in vacuum extraction: A web-based survey among doctors in Sweden

Victoria Ankarcrona [a,*], Susanne Hesselman [b], Helena Kopp Kallner [a], Sophia Brismar Wendel [a]

Implementation in clinical practice

Episiotomy in vacuum births



OASI in vacuum births

Approval from VR – and the next project…

The Aortic Compression Trial

- Can routine aortic compression reduce blood loss in cesarean section?

Funding granted!

Planning **1.2 MSEK** 2022-2023

Clinical therapy **19.8 MSEK** 2025-2028

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